

Introduction to Economics

Class 5 — Competitive markets, demand and supply

Pavel Kocourek

Universitat de Barcelona · Group A6

5 October 2026

slide 1 of 29

Slide 1 Title

Class 5 opens Block 2: markets. Block 1 asked what is possible and what it costs; from today the question is what sets the price. The first hour is the buyers' side, demand, built around a real price cut on the Barcelona metro; the second hour mirrors it on the sellers' side, supply. One curve at a time today, the two together on Thursday.

Quiz 3

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

quiz page, not numbered

Quiz Start of class, on paper

A short quiz on paper opened the class: circle the letter of your answer in the block at the top of the sheet, write your name on the block, tear the strip off along the dashed line and hand in only the strip.

Plan for today

10:00	Quiz 3
10:08	Block 1, one more look
10:13	Block 2 begins: the metro's price cut
10:16	A competitive market
10:19	Demand: a curve, a move along it, a shift
10:50	Break
11:00	Project: the sign-up sheet
11:03	Supply: a curve, a move along it, a shift
11:31	Both curves on one picture
11:40	Before Thursday

Slide 2 Plan for today

The class at a glance. First a second look at two Block 1 ideas; then the metro's price cut, what a competitive market is, and demand: a curve, a move along it, a shift. After the break the project sign-up sheet, then supply built the same way, and finally both curves on one picture.

Two halves of fifty minutes with a ten-minute break between them. The results of Thursday's quiz are on the course page, under Class 4.

Block 1, one more look: same cost on both sides

Pau bakes 4 croissants or 2 ensaïmadas an hour; Júlia 8 or 4. Which deal makes both better off?

- A) Júlia bakes everything: she is faster at both
- B) Pau sells Júlia ensaïmadas at 1 croissant each
- C) Pau sells Júlia ensaïmadas at 3 croissants each
- D) None: an ensaïmada costs both 2 croissants ✓**

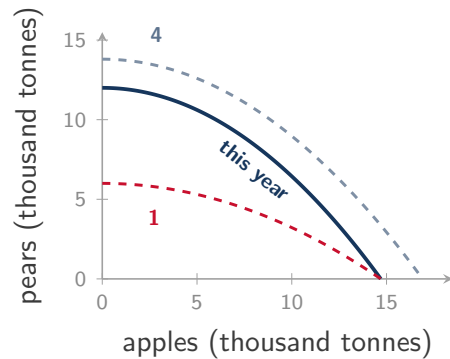
At 1, Pau gives away 2 croissants' worth for 1. At 3, Júlia pays 3 for what costs her 2. At 2, neither gains. Faster at both is not a reason to trade; a different cost is.

Slide 3 Block 1, one more look: same cost on both sides

Pau bakes 4 croissants or 2 ensaïmadas an hour, Júlia 8 or 4. For both of them an ensaïmada costs two croissants: the same opportunity cost on both sides, so no deal makes both better off (D). At one croissant per ensaïmada Pau gives away what cost him two; at three Júlia pays three for what she could bake for two; at exactly two the trade can happen, but neither gains.

A is the trap: Júlia is faster at both, an absolute advantage, and that is not a reason to trade. Trade needs a difference in opportunity costs, as in Class 3's example: three coffees per bocadillo on one side, one on the other.

Block 1, one more look: what moves the frontier?



- 1 A storm flattens half the pear orchards.
moves it: the pear end comes in
- 2 The world price of apples doubles.
no move: nothing it can make has changed
- 3 The island grows more pears, fewer apples.
a move along the frontier
- 4 New irrigation helps both crops.
moves it out, both ends

Resources and technology move the frontier; prices move you along it — today's rule, twice.

Slide 4 Block 1, one more look: what moves the frontier?

Thursday's island, apples and pears, and four events. A storm that flattens half the pear orchards moves the frontier: the pear end comes in and the apple end stays (dashed curve 1). A doubling of the world price of apples moves nothing: a price changes what the island chooses, not what it can make. Growing more pears and fewer apples is a move along the frontier, from one point to another on the same curve. New irrigation that helps both crops moves the whole frontier out (dashed curve 4).

The rule, and the bridge into today: resources and technology move the frontier; prices and choices move you along it. The same rule returns twice today, on two new curves.

Block 2: markets

What sets the price: demand and supply

Slide 5 Block 2: markets

Block 2 begins. Block 1 asked what is possible and what it costs; Block 2 asks what sets the price, in the reading's model: a competitive market with a demand side and a supply side.

September 2022: a month of metro for 20 euros, not 40



T-usual, one zone, from 1 September 2022

What did people do?

ATM (Autoritat del Transport Metropolità), fares from 1 September 2022: the T-usual at half price.

Slide 6 September 2022: a month of metro for 20 euros, not 40

The hook of the class. From 1 September 2022 the T-usual, the monthly pass for one zone, cost 20 euros instead of 40: a 30 % discount paid by the state in its energy-crisis package, topped up to 50 % by the Generalitat, the AMB and the city council; the half price stayed through 2023. One seller, and a price cut by decree. What did people do?

2023: a record 1,087 million journeys, 17 % more than 2022

1,087 million

public transport journeys in 2023
+17 % on 2022 · +3 % on 2019

53.5 % → 17 %

the ten-ride ticket's share
of journeys, 2019 to 2023
(T-10, then T-casual)

Price halved, journeys up. All the price's doing? Back to it at the end of the hour.

ATM, press release of 25 January 2024 (journeys in the integrated area; shares of journeys on integrated tickets).

Slide 7 2023: a record 1,087 million journeys, 17 % more than 2022

The reveal. The ATM, which runs the integrated fares, closed 2023 with a record 1,087 million journeys in the Barcelona area, 17 % more than in 2022 and 3 % above 2019, the last year before Covid. Inside the total the tickets swapped places: the ten-ride ticket (the T-10, replaced in January 2020 by the dearer, one-person T-casual) carried 53.5 % of journeys on integrated tickets in 2019 and only 17 % in 2023; the T-usual is now the most used.

Price halved, journeys up. Is all of that the price's doing? The question stays open until slide 16.

A competitive market: many buyers, many sellers, one price

- One good, the same from every seller
- So many buyers and sellers that no one can move the price
- Everyone takes the price as given

Mercabarna's tomatoes: close. The metro: one seller — its buyers still follow a demand curve.

Slide 8 A competitive market: many buyers, many sellers, one price

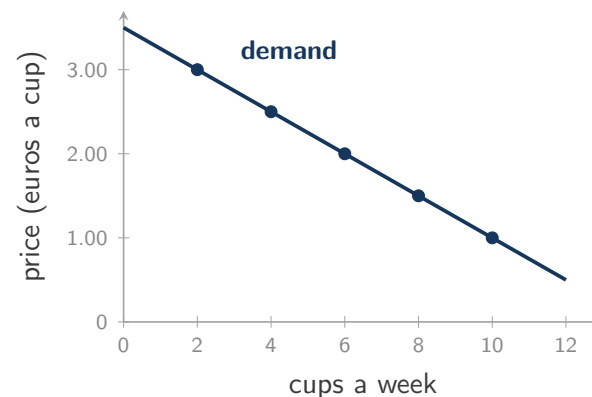
A market is the buyers and sellers of one good. A competitive market is one where the good is the same from every seller and there are so many buyers and sellers that none of them can move the price: everyone takes it as given. Tomatoes at Mercabarna, the wholesale market in the Zona Franca, come close.

The metro is not a competitive market: one seller, and a price set by decree. It opens the class because on the buyers' side it behaves exactly as the model says, more journeys at a lower price. The seller's side is where it differs, and we leave that alone.

Demand: at each price, how many cups Mireia buys

price (€ a cup)	cups a week
1.00	10
1.50	8
2.00	6
2.50	4
3.00	2

Mireia's demand schedule



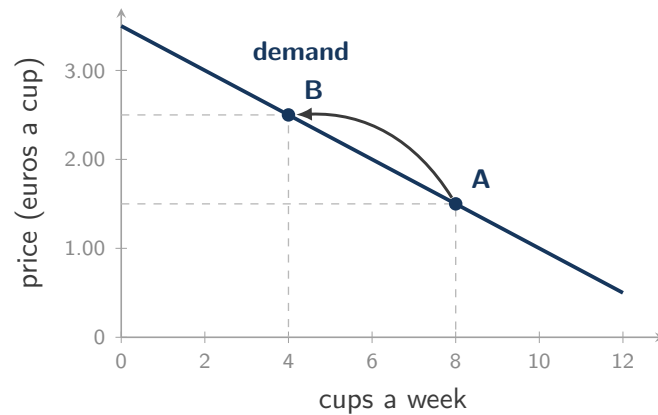
Law of demand: other things equal, the price rises, the quantity demanded falls.

Slide 9 Demand: at each price, how many cups Mireia buys

One buyer. Mireia's demand schedule lists, for each price of a café amb llet at the faculty bar, how many cups a week she buys: 10 at 1 euro, 8 at 1.50, down to 2 at 3 euros. Plotted and joined, the points are her demand curve. Price goes on the vertical axis and quantity on the horizontal, by convention.

The quantity demanded is the amount buyers are willing and able to buy at a given price. The law of demand: other things equal, when the price rises the quantity demanded falls, so the curve slopes down. "Other things equal" is the phrase to underline: the schedule holds Mireia's income, her tastes and every other price fixed.

A dearer cup: fewer cups, the same curve



A to B: the quantity demanded fell from 8 to 4. The curve — demand — did not move.

Pavel Kocourek

Introduction to Economics · Class 5

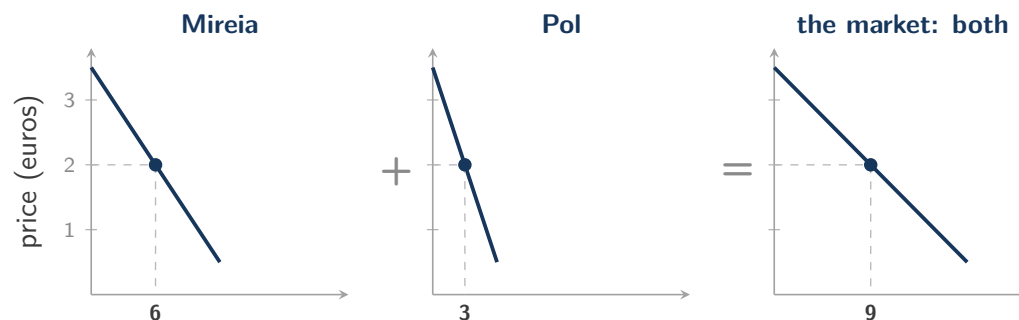
10

slide 10 of 29

Slide 10 A dearer cup: fewer cups, the same curve

The bar raises the price from 1.50 to 2.50. Mireia moves from A to B along her curve: 8 cups a week become 4. Two sentences to keep: the quantity demanded fell; demand, the curve, did not change. A price change never moves the curve, because the curve already contains every price.

Market demand: at each price, add the buyers up



At 2 euros, Mireia 6 + Pol 3 = 9 a week: the buyers' curves added sideways.

Pavel Kocourek

Introduction to Economics · Class 5

11

slide 11 of 29

Slide 11 Market demand: at each price, add the buyers up

The market is all the buyers. At each price, add up what each of them wants: at 2 euros Mireia wants six cups and Pol three, so the market wants nine, and the same at every other price. On the picture that is adding the curves sideways, horizontally, never vertically; the dashed guide at 2 euros shows it. In the reading's words, the market demand curve is the horizontal sum of the individual demand curves.

The market curve slopes down for the same reason as each buyer's, and for one more: at a lower price some people buy who bought nothing before.

In pairs: add up the market

Three buyers of cups a week. Fill the last column; then the question.

price	Ona	Biel	Roser	the market
1.00	6	4	5	15 cups
1.50	5	3	4	12 cups
2.00	4	2	3	9 cups
2.50	3	1	2	6 cups

The bar raises the price from 1.50 to 2.00: a move along the market curve, or a shift of it?

Along it: 12 cups to 9. Nothing but the price changed, so demand did not.

Slide 12 In pairs: add up the market

Three buyers' schedules. The market column is the sum across each row: 15 cups at 1 euro, 12 at 1.50, 9 at 2, 6 at 2.50. Then the question: the bar raises the price from 1.50 to 2.00. That is a move along the market curve, from 12 cups to 9: nothing but the price changed, so demand did not.

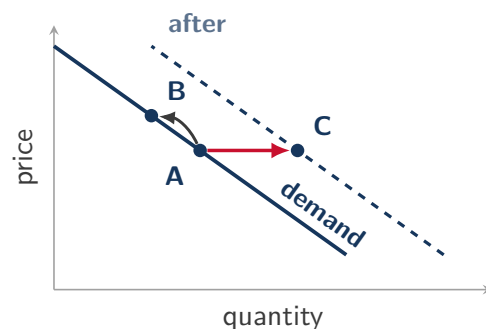
The trap is to say "demand fell". It did not; the quantity demanded fell.

Slide 13 Price moves you along the curve. All else moves the curve.

The slide of the day. From A to B the price rose and we slid along the curve. From A to C, at the same price, the market wants more: the whole curve has moved to the right, which only happens when something other than the price changed.

Along the curve, the quantity demanded changes. A shift, and demand changes. A shift to the right means more wanted at every price; to the left, less at every price.

Price moves you along the curve. All else moves the curve.



A → B

a move along the curve:
the price changed

A → C

the curve moved:
something else changed

Along the curve: the **quantity demanded** changes. A shift: **demand** changes.

What moves the demand curve

- **Income** rises: normal goods up, inferior goods down
- Price of a **substitute**: tea dearer, more coffee
- Price of a **complement**: croissants dearer, less coffee
- **Tastes** and **expectations**
- The **number of buyers**

Right: more wanted at every price. Left: less at every price.

Slide 14 What moves the demand curve

What shifts the demand curve: anything that matters to buyers other than the good's own price. Income: when it rises, demand for a normal good shifts right and demand for an inferior good shifts left (the cheapest instant noodles, once you find a job). The price of a substitute, a good you buy instead: tea dearer, more coffee, a shift right. The price of a complement, a good you use together with it: croissants dearer, less coffee, a shift left.

Tastes, and advertising, which works on them. Expectations: if you expect the price to rise next month, you buy now. And the number of buyers: more people on the campus, more cups at every price.

In pairs: demand for metro journeys — along the curve, or a shift?

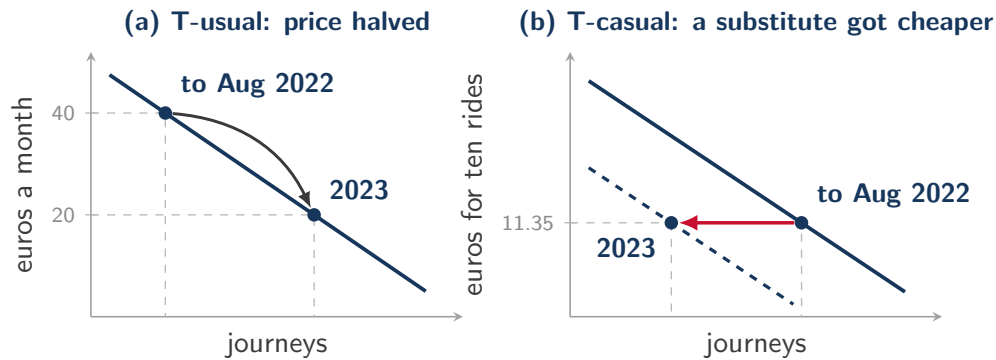
- 1 The T-usual drops from 40 to 20 euros.
along the curve, down and to the right
- 2 Offices reopen after Covid.
shift right: more buyers
- 3 Petrol gets dearer.
shift right: driving, a substitute, got dearer
- 4 Taxis get cheaper.
shift left: a substitute got cheaper
- 5 Cycling to work becomes the fashion.
shift left: tastes changed

Slide 15 In pairs: demand for metro journeys

Five events, with the curve showing journeys at each price of the pass. The test from the previous slides: did the metro's own price change, or something else? Only item 1 moves the price, so only item 1 is a move along the curve; the other four shift it, in the directions on the slide.

Two traps: reading "more journeys" as if the price had fallen (item 2 is a shift, more buyers), and reading a fashion for the substitute as a fashion for the metro (item 5 shifts left, not right). Item 3 is the subtle one: petrol is not the metro's substitute, driving is, and it got dearer.

One price cut: along one curve, a shift of the other



2023: the T-usual first, 37 % of journeys; the T-casual 17 %. Buyers moved to the substitute.

ATM, press release of 25 January 2024; fares: ATM, from 1 September 2022. The curves are sketches, not data.

Pavel Kocourek

Introduction to Economics · Class 5

16

slide 16 of 29

Slide 16 One price cut: along one curve, a shift of the other

The hook's question answered. Panel (a), the T-usual: its own price halved, so its buyers slid along their curve, from 40 euros to 20 (slide 15, item 1). Panel (b), the T-casual: by 2023 its price was back where it had been before the cut, 11.35 euros for ten rides, but a substitute got cheaper, so its curve shifted left. The T-usual became the most used ticket, 37 % of journeys on integrated tickets; the T-casual 17 %.

The 17 % rise in total journeys mixes the two: the price cut moved buyers along the curve; everything else, from people back in offices after Covid to more tourists, moved the curve. Only the first part is the price's doing; separating the two needs both curves and a method: Thursday's class. The curves are sketches, not data.

Slide 17 Break

Ten minutes. Sit with your project group afterwards: the sign-up sheet for the idea meetings goes round right after the break.

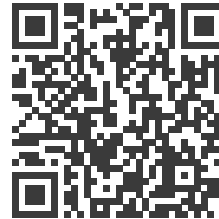
Break

Back at 11:00

slide 17 of 29

Project: sign up for your idea meeting

- The sheet goes round now: one slot per group, six minutes
- Thursday 8 October, 12:00–13:00
- The day before your slot: your idea in one paragraph, by e-mail
- Full brief on the course page



course page

Slide 18 Project: sign up for your idea meeting

The sign-up sheet for the idea meetings: one slot per group, six minutes each, on Thursday 8 October straight after class. The day before your slot, e-mail your idea in one paragraph, so the six minutes go on the question rather than on explaining it. The project brief on the course page (QR code) says what makes a good question: specific, local, something you noticed yourselves.

Groups were formed in Class 4; anyone still without one joins a group now. The topic is fixed on Thursday 15 October; the poster session is on Thursday 5 November.

Slide 19 Supply

Supply: the sellers' side of the same market. The same three questions, mirrored: how much does one seller offer at each price, how do we add the sellers up, and what moves the curve.

Supply

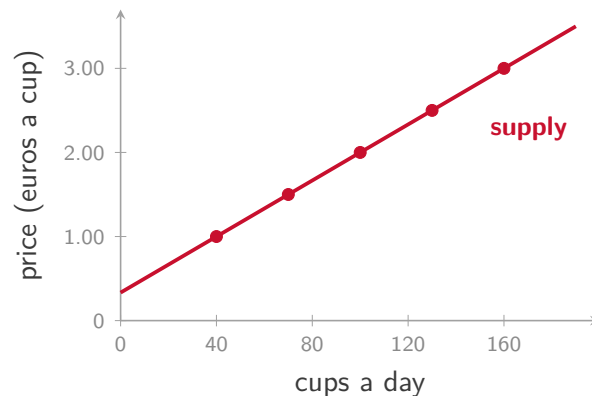
The sellers' side of the same market

Supply: at each price, how many cups Bar Nou makes

price (€ a cup)	cups a day
1.00	40
1.50	70
2.00	100
2.50	130
3.00	160

Bar Nou's supply schedule

Law of supply: other things equal, the price rises, the quantity supplied rises.

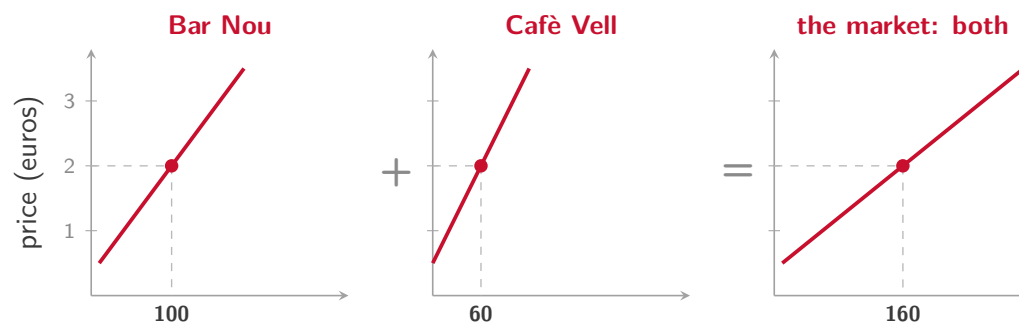


Slide 20 Supply: at each price, how many cups Bar Nou makes

One seller. Bar Nou's supply schedule lists, for each price, how many cups of café amb llet a day it makes: 40 at 1 euro, 70 at 1.50, up to 160 at 3 euros. Joined, the points are its supply curve. The quantity supplied is the amount sellers are willing and able to sell at a given price. The law of supply: other things equal, when the price rises the quantity supplied rises, so the curve slopes up.

Why up: at a higher price it pays to open longer, put a second person on the machine and serve the slow customer too; each extra cup costs a little more to make, and a higher price covers it. Supply curves are red and demand curves navy throughout the course.

Market supply: at each price, add the sellers up



At 2 euros, Bar Nou 100 + Cafè Vell 60 = 160 a day: the sellers' curves added sideways.

Slide 21 Market supply: at each price, add the sellers up

The market is all the sellers. At each price, add up what each of them offers: at 2 euros Bar Nou makes 100 and Cafè Vell 60, so the market offers 160 cups a day. The same move as for the buyers: the curves are added sideways, never vertically. In the reading's words, the market supply curve is the horizontal sum of the individual supply curves.

Below 50 cents only Bar Nou sells, so the market curve has a kink there; the three panels are drawn from 50 cents up.

In pairs: add up the sellers

Two sellers of cups a day. Fill the last column; then the two questions.

price	Forn Gran	Bar Petit	the market
1.00	30	10	40 cups
1.50	50	20	70 cups
2.00	70	30	100 cups
2.50	90	40	130 cups

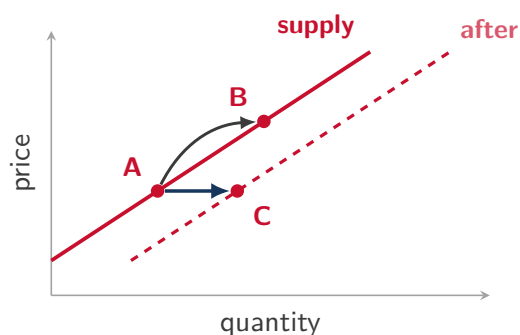
- 1 The price rises from 1.50 to 2.00 — along the curve or a shift?
Along it: 70 cups to 100. The curve did not move.
- 2 Bar Petit closes — along the curve or a shift?
A shift left: fewer sellers. The market loses Bar Petit's column.

Slide 22 In pairs: add up the sellers

Two sellers; the market column is the row sum: 40 cups at 1 euro, 70 at 1.50, 100 at 2, 130 at 2.50. Question 1: the price rises from 1.50 to 2.00, a move along the market curve from 70 cups to 100; the curve did not move. Question 2: Bar Petit closes. That is a shift of the market curve to the left, fewer sellers: at every price the market now offers only Forn Gran's column.

The trap: a seller leaving is not a price change, so it cannot be a move along the curve.

Price moves you along the supply curve. All else moves the curve.



A → B
a move along the curve:
the price changed

A → C
the curve moved:
something else changed

Along the curve: the **quantity supplied** changes. A shift: **supply** changes.

Slide 23 Price moves you along the supply curve. All else moves the curve.

The slide of the day, mirrored. From A to B the price rose and the seller slid up along the supply curve. From A to C, at the same price, the market offers more: the curve has moved to the right, because something other than the price changed.

Along the curve, the quantity supplied changes. A shift, and supply changes. Right means more offered at every price; left, less. "All else" is wider than costs: the number of sellers and expectations move supply too.

What moves the supply curve

- **Input prices:** beans, milk, wages, rent
- **Technology:** more cups an hour
- **Expectations** about the price
- The **number of sellers**

Right: more offered at every price. Left: less at every price.

Slide 24 What moves the supply curve

What shifts the supply curve: anything that matters to sellers other than the good's own price. Input prices, such as beans, milk, wages and rent: dearer inputs mean less offered at every price, a shift left. Technology, such as a machine that makes two cups a minute: a shift right. Expectations about the price: a seller who expects a higher price next month and can store the good holds back today, so today's supply shifts left; a firm that expects higher prices builds capacity, and later supply shifts right. The number of sellers: more bars on the street, a shift right.

The reading adds the weather and other natural factors, and how profitable the other goods a firm could make instead are.

In pairs: the street's supply of coffee — along it, or a shift?

- 1 Coffee beans double in price.
shift left: an input got dearer
- 2 A new machine makes two cups a minute.
shift right: a better machine
- 3 The price of a cup rises to 2 euros.
along the curve, up and to the right
- 4 Three new bars open on the street.
shift right: more sellers
- 5 Baristas' wages rise.
shift left: wages are an input

Slide 25 In pairs: the street's supply of coffee

The street's supply of coffee is the market curve of slides 21 and 22, all the cafès together. The test: did the price of a cup change, or something the sellers care about? Only item 3 moves the price, so only item 3 is a move along the curve; the other four shift it, in the directions on the slide.

The trap: a dearer bean is a cost, not the price of the cup, so item 1 is a shift, not a move along the curve. Item 4 is about the street because one café's own curve does not move when neighbours open.

Two pairs of words that are not the same

what changed	on the picture	say
the price	along the curve	quantity demanded · quantity supplied
anything else	the curve shifts	demand · supply

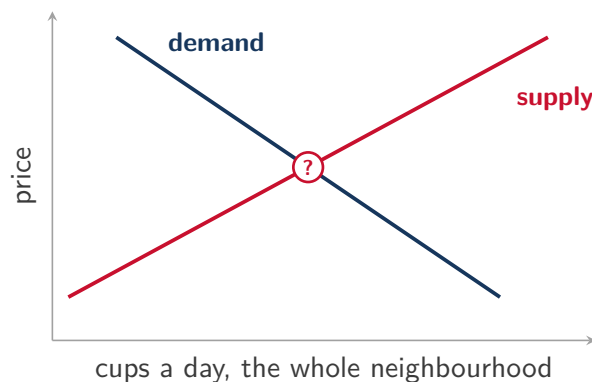
“Demand rose” means the curve moved. “Quantity demanded rose” means the price fell.

Slide 26 Two pairs of words that are not the same

The table says it. When the price changes, you move along the curve, and the words are quantity demanded or quantity supplied. When anything else changes, the curve shifts, and the words are demand or supply. “Anything else” means anything else that matters to buyers (for demand) or to sellers (for supply): an input price moves supply, not demand.

“Demand rose” means the curve moved. “Quantity demanded rose” means the price fell. Midterm 1 marks this wording.

Demand and supply on one picture: where they cross, Thursday



Thursday: what happens at the crossing — and what moves it.

Slide 27 Demand and supply on one picture

Both curves on one picture: the whole neighbourhood’s buyers (navy) and sellers (red) of càfé amb llet. They cross once. What happens at the crossing, why the market goes there, and what happens to the price when one curve moves, is Thursday’s class: every shift from today’s two sorting exercises becomes a price change there.

Price moves you along a curve. Everything else moves the curve.

Demand and supply: anything but the good's own price.

The frontier: resources and technology move it.

slide 28 of 29

Slide 28 In one line

The class in one line: price moves you along a curve; everything else moves the curve. For demand and supply, “everything else” is anything but the good's own price. The frontier of Block 1 has its own version of the rule: resources and technology move it, the island's choices move you along it, and prices are not on its axes.

Before Thursday 8 October

- Read Mankiw & Taylor, Ch. 3, pp. 44–52 — supply and demand together, prices as signals, changes in equilibrium
- Practise Block 1 in Sextant: the units for Classes 2 and 3
- Project: idea meetings Thursday, 12:00–13:00; your idea by e-mail the day before your slot
- Thursday 8 October — equilibrium: where the two curves meet



pkocourek@ub.edu · pkocourek.com/teaching/intro-economics · pkocourek.com/sextant

Slide 29 Before Thursday 8 October

Before Thursday: read Mankiw & Taylor chapter 3, pages 44 to 52, on supply and demand together, prices as signals and changes in equilibrium. Practise Block 1 in Sextant with the units for Classes 2 and 3. Project: the idea meetings are on Thursday straight after class; your idea in one paragraph by e-mail the day before your slot.

Thursday: equilibrium, where the two curves meet.

pkocourek.com/teaching/intro-economics · pkocourek@ub.edu